PA3 — Augmented Reality Website with Hit-Testing

Student ID: 241629
Variant: Standing Desk
Device: Samsung Galaxy A35 5G

E1 — Viewer pose and hit testing

Prediction: As the phone moves, the viewer pose changes with the device position and orientation. A hit test uses the viewer's pose to find a point on a real surface.

Result: The AR session became active, and the application displayed reticle feedback on real surfaces. This shows that the viewer tracking and hit-test flow were running during the AR session.

Explanation: The viewer pose describes the device's position and orientation in the AR reference space. The hit-test result supplies a pose for the detected surface, which the application can use to position virtual content.

E2 — Anchors and reference spaces

Prediction: An anchor should keep virtual content associated with a real-world position as the viewer moves. The reference space determines how positions are represented during the session.

Result: The screenshots show an active AR session with "Anchors: on." The application detected the tabletop while anchors were enabled.

Explanation: Anchors provide a stable reference for virtual content in the tracked environment. A local reference space represents positions relative to the user's environment, while a local-floor reference space uses the floor as its origin.

E3 — Hit-test reticle and surface angle

Prediction: A sufficiently horizontal surface should produce a valid green reticle. A surface that exceeds the allowed tilt should produce a red reticle and block placement.

Result: The green reticle is labelled "valid (0.0°, limit 8°)." The red reticle is labelled "too tilted (90.0°, limit 8°)." In both cases, the AR session is active.

Explanation: The horizontal surface is within the 8° limit, so the reticle indicates that placement is allowed. The vertical surface exceeds the limit, so the reticle indicates that placement is blocked. Figures 1–4 show these two states.

E4 — Surface detection

Prediction: The AR system should detect a suitable flat tabletop and display a valid reticle over it.

Result: Three screenshots show a green, valid reticle on the white tabletop. The tabletop was detected as a suitable placement surface in these views.

Explanation: The valid reticle provides visual feedback that the hit test found a surface suitable for placement. The red reticle shown in Figure 2 provides a contrasting example of a surface rejected because of its angle.

E5 — Virtual standing desk scale

Prediction: The standing desk should appear at a plausible scale when its model is placed on the detected tabletop.

Result: The AR scene shows the reticle positioned on the tabletop, providing a surface reference for placing the virtual desk.

Explanation: The model's geometry determines the virtual object's dimensions, and the hit-test pose determines its position and orientation on the surface. For a numerical real-world scale comparison, the model and physical tabletop must be measured using the same units.

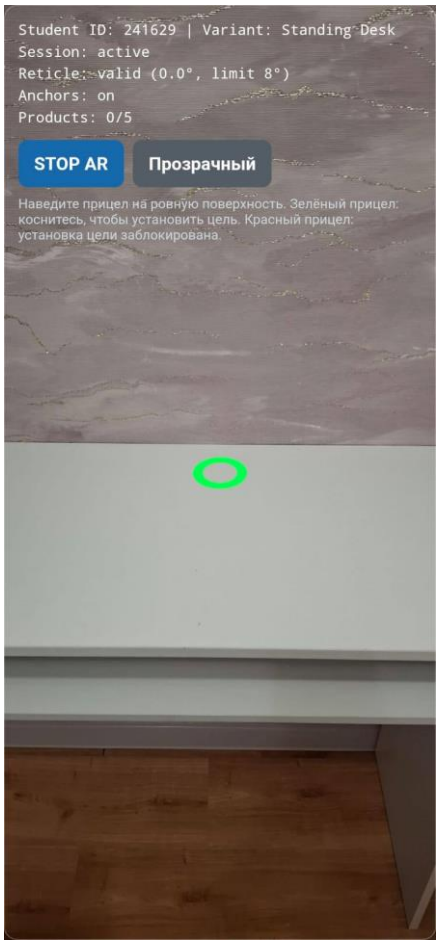
E6 — Evaluation and improvements

The reticle gives immediate feedback about whether a surface can be used for placement. The green state indicates a valid horizontal surface, while the red state blocks placement on a steep surface. The interface also displays the session state, anchor state, and product count. Further improvements could include a clearer placement confirmation and additional visual guidance while the application searches for surfaces. The application uses AR tracking and hit-test information to position virtual content.

Screenshot captions




Student ID: 241629 | Variant: Standing Desk
Session: active
Reticle: too tilted (90.0°, limit 8°)
Anchors: on
Products: 0/5
STOP AR
Clear
Aim at a flat surface. Green reticle: tap to place. Red reticle: placement is blocked.


Student ID: 241629 | Variant: Standing Desk
Session: active
Reticle: valid (0.0°, limit 8°)
Anchors: on
Products: 0/5
STOP AR
Прозрачный
Наведите прицел на ровную поверхность. Зелёный прицел: коснитесь, чтобы установить цель. Красный прицел: установка цели заблокирована.

Development log

A desk-foot orientation bug made the feet extend along the wrong axis.
I found it by inspecting the rendered model and comparing the feet with the intended desk layout.
I corrected the dimensions so the feet extend along the desk’s depth.
The fix is included in commit 2912f6057094a026ebe6b34da1b7d7a4e09bb90f.